ECET
250E

Linear Circuits 1

Week 3 | Chapter 3: Methods of Analysis

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Overview

With Ohm's and Kirchhoff's law established, they may now be applied to circuit analysis.

Two techniques will be presented in this chapter:

- **Nodal** analysis, which is based on Kichhoff current law (**KCL**).
- **Mesh** analysis, which is based on Kichhoff voltage law (**KVL**).

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Nodal Analysis

- If instead of focusing on the voltages of the circuit elements, one looks at the **voltages at the nodes** of the circuit, the number of simultaneous equations to solve for can be reduced.
- Given a circuit with **n** nodes, without voltage sources, the nodal analysis is accomplished via three steps:

  1. Select **a** node as the **reference** node. Assign voltages $v_1, v_2, \ldots v_n$ to the remaining *n-1* nodes, voltages are relative to the reference node.
  2. Apply **KCL** to each of the ***n − 1* non-reference nodes**. Use Ohm's law to express the branch currents in terms of node voltages.
  3. Solve the resulting $n - 1$ simultaneous equations to obtain the unknown node voltages.

- The reference, or datum, node is commonly referred to as the ground since its voltage is by default zero.

Nodal analysis is also known as the *node-voltage method*.

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Applying Nodal Analysis

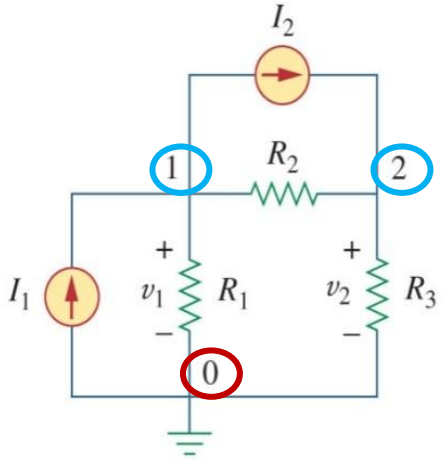
- Let's apply nodal analysis to this circuit to see how it works.
- This circuit has a node that is designed as ground. We will use that as the reference node (**node 0**).
- The remaining two nodes are designed **1** and **2** and assigned voltages v_1 and v_2.
- Now apply **KCL** to each node:
- At node 1. $I_1 = I_2 + i_1 + i_2$
- At node 2. $I_2 + i_2 = i_3$

Current flows from a higher potential to a lower potential in a resistor.

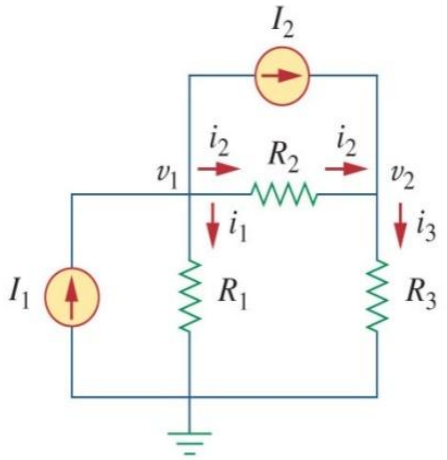
We can express this principle as

$$i = \frac{v_{higher} - v_{lower}}{R}$$

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(a)



(b)

Apply Nodal Analysis II

- We can now use OHM's law to express the unknown currents ***i*₁, *i*₂,** *and* ***i*₃** in terms of node voltages.
- In doing so, keep in mind that current flows from high potential to low.
- From this we get:

$$I_1 = I_2 + i_1 + i_2$$
$$I_2 + i_2 = i_3$$

$$i_1 = \frac{v_1 - 0}{R_1} \quad \text{or} \quad i_1 = G_1 v_1$$

$$i_2 = \frac{v_1 - v_2}{R_2} \quad \text{or} \quad i_2 = G_2 (v_1 - v_2)$$

$$i_3 = \frac{v_2 - 0}{R_3} \quad \text{or} \quad i_3 = G_3 v_2$$

Substituting back into the node equations →

$$I_1 = I_2 + \frac{v_1}{R_1} + \frac{v_1 - v_2}{R_2}$$

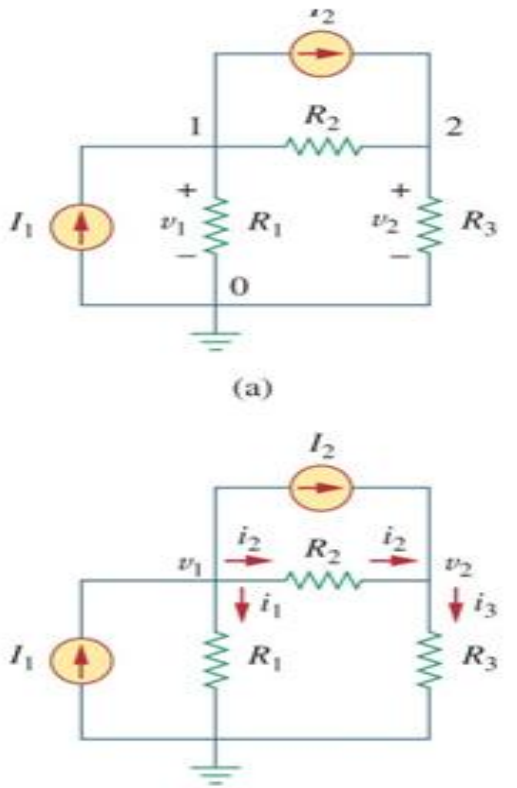
$$I_2 + \frac{v_1 - v_2}{R_2} = \frac{v_2}{R_3}$$

or

$$I_1 = I_2 + G_1 v_1 + G_2 (v_1 - v_2)$$

$$I_2 + G_2 (v_1 - v_2) = G_3 v_2$$

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- The last step is to solve the system of equations.

Apply Nodal Analysis

Example 3.1

Calculate the node voltages in the circuit shown in Fig. 3.3(a).

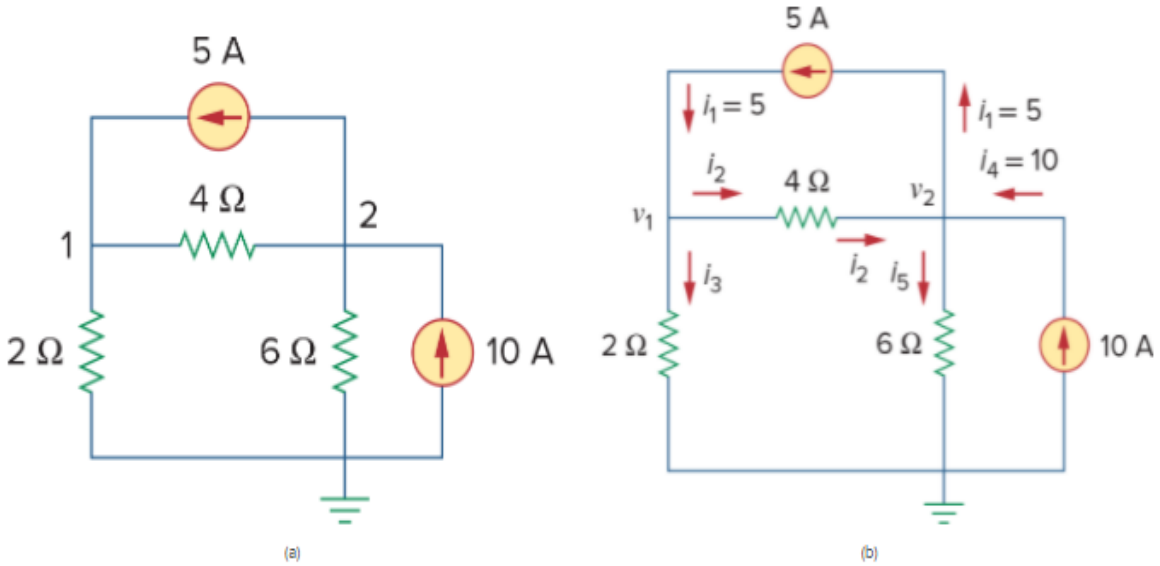


Figure 3.3
For Example 3.1: (a) original circuit, (b) circuit for analysis.

Solution:

Consider Fig. 3.3(b), where the circuit in Fig. 3.3(a) has been prepared for nodal analysis. Notice how the currents are selected for the application of KCL. Except for the branches with current sources, the labeling of the currents is arbitrary but consistent. (By consistent, we mean that if, for example, we assume that i_2 enters the 4-Ω resistor from the left-hand side, i_2 must leave the resistor from the right-hand side.) The reference node is selected, and the node voltages v_1 and v_2 are now to be determined.

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At node 1, applying KCL and Ohm's law gives

$$i_1 = i_2 + i_3 \quad \Rightarrow \quad 5 = \frac{v_1 - v_2}{4} + \frac{v_1 - 0}{2}$$

Multiplying each term in the last equation by 4, we obtain

$$20 = v_1 - v_2 + 2v_1$$

or

$$3v_1 - v_2 = 20 \tag{3.1.1}$$

At node 2, we do the same thing and get

$$i_2 + i_4 = i_1 + i_5 \quad \Rightarrow \quad \frac{v_1 - v_2}{4} + 10 = 5 + \frac{v_2 - 0}{6}$$

Multiplying each term by 12 results in

$$3v_1 - 3v_2 + 120 = 60 + 2v_2$$

or

$$-3v_1 + 5v_2 = 60 \tag{3.1.2}$$

Now we have two simultaneous Eqs. (3.1.1) and (3.1.2). We can solve the equations using any method and obtain the values of v_1 and v_2.

■ METHOD 1 Using the elimination technique, we add Eqs. (3.1.1) and (3.1.2).

$$4v_2 = 80 \quad \Rightarrow \quad v_2 = 20 \text{ V}$$

Substituting $v_2 = 20$ in Eq. (3.1.1) gives

$$3v_1 - 20 = 20 \quad \Rightarrow \quad v_1 = \frac{40}{3} = 13.333 \text{ V}$$

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■ METHOD 2 To use Cramer's rule, we need to put Eqs. (3.1.1) and (3.1.2) in matrix form as

$$\begin{bmatrix} 3 & -1 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 20 \\ 60 \end{bmatrix} \tag{3.1.3}$$

The determinant of the matrix is

$$\Delta = \begin{bmatrix} 3 & -1 \\ -3 & 5 \end{bmatrix} = 15 - 3 = 12$$

We now obtain v_1 and v_2 as

$$v_1 = \frac{\Delta_1}{\Delta} = \frac{\begin{vmatrix} 20 & -1 \\ 60 & 5 \end{vmatrix}}{\Delta} = \frac{100 + 60}{12} = 13.333 \text{ V}$$

$$v_2 = \frac{\Delta_2}{\Delta} = \frac{\begin{vmatrix} 3 & 20 \\ -3 & 60 \end{vmatrix}}{\Delta} = \frac{180 + 60}{12} = 20 \text{ V}$$

giving us the same result as did the elimination method.

If we need the currents, we can easily calculate them from the values of the nodal voltages.

$$i_1 = 5 \text{ A}, \quad i_2 = \frac{v_1 - v_2}{4} = -1.6668 \text{ A}, \quad i_3 = \frac{v_1}{2} = 6.666 \text{ A}$$

$$i_4 = 10 \text{ A}, \quad i_5 = \frac{v_2}{6} = 3.333 \text{ A}$$

The fact that i_2 is negative shows that the current flows in the direction opposite to the one assumed.

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Practice Problem 3.1

Obtain the node voltages in the circuit of Fig. 3.4.

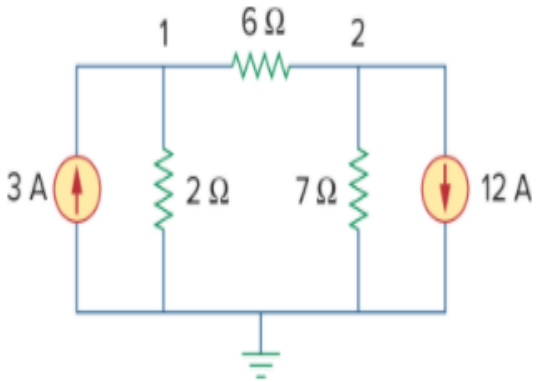
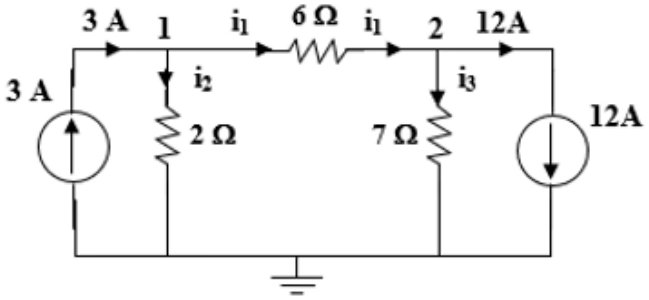


Figure 3.4
For Practice Prob. 3.1.

Answer: $v_1 = -6$ V, $v_2 = -42$ V.

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P.P.3.1



At node 1,

$$-3 + i_1 + i_2 = 0 \text{ or } \frac{v_1 - v_2}{6} + \frac{v_1 - 0}{2} = 3$$

or $4v_1 - v_2 = 18$ (1)

At node 2,

$$-i_1 + i_3 + 12 = 0 \text{ or } i_1 = 12 + i_3 \text{ or } \frac{v_1 - v_2}{6} = 12 + \frac{v_2 - 0}{7}$$

or $7v_1 - 13v_2 = 504$ (2)

Solving (1) and (2) gives

$v_1 = \mathbf{-6\ V}$, $v_2 = \mathbf{-42\ V}$

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Example 3.2

Determine the voltages at the nodes in Fig. 3.5(a).

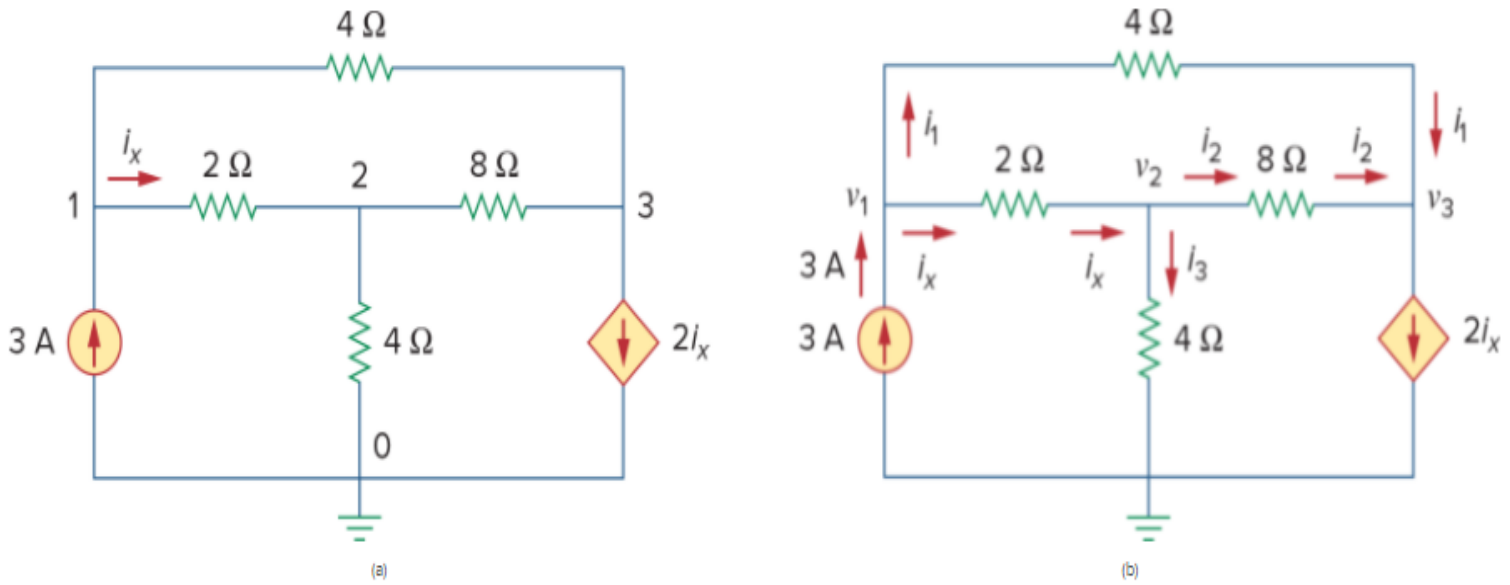


Figure 3.5
For Example 3.2: (a) original circuit, (b) circuit for analysis.

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Solution:

The circuit in this example has three nonreference nodes, unlike the previous example which has two nonreference nodes. We assign voltages to the three nodes as shown in Fig. 3.5(b) and label the currents.

At node 1,

$$3 = i_1 + i_x \quad \Rightarrow \quad 3 = \frac{v_1 - v_3}{4} + \frac{v_1 - v_2}{2}$$

Multiplying by 4 and rearranging terms, we get

$$3v_1 - 2v_2 - v_3 = 12 \tag{3.2.1}$$

At node 2,

$$i_x = i_2 + i_3 \quad \Rightarrow \quad \frac{v_1 - v_2}{2} = \frac{v_2 - v_3}{8} + \frac{v_2 - 0}{4}$$

Multiplying by 8 and rearranging terms, we get

$$-4v_1 + 7v_2 - v_3 = 0 \tag{3.2.2}$$

At node 3,

$$i_1 + i_2 = 2i_x \quad \Rightarrow \quad \frac{v_1 - v_3}{4} + \frac{v_2 - v_3}{8} = \frac{2(v_1 - v_2)}{2}$$

Multiplying by 8, rearranging terms, and dividing by 3, we get

$$2v_1 - 3v_2 + v_3 = 0 \tag{3.2.3}$$

We have three simultaneous equations to solve to get the node voltages v_1, v_2, and v_3. We shall solve the equations in two ways.

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■ METHOD 1 Using the elimination technique, we add Eqs. (3.2.1) and (3.2.3).

$$5v_1 - 5v_2 = 12$$

or

$$v_1 - v_2 = \frac{12}{5} = 2.4 \tag{3.2.4}$$

Adding Eqs. (3.2.2) and (3.2.3) gives

$$-2v_1 + 4v_2 = 0 \quad \Rightarrow \quad v_1 = 2v_2 \tag{3.2.5}$$

Substituting Eq. (3.2.5) into Eq. (3.2.4) yields

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$$2v_2 - v_2 = 2.4 \quad \Rightarrow \quad v_2 = 2.4. \quad v_1 = 2v_2 = 4.8 \text{ V}$$

From Eq. (3.2.3), we get

$$v_3 = 3v_2 - 2v_1 = 3v_2 - 4v_2 = -v_2 = -2.4 \text{ V}$$

Thus,

$$v_1 = 4.8 \text{ V}, \quad v_2 = 2.4 \text{ V}. \quad v_3 = -2.4 \text{ V}$$

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■ METHOD 2 To use Cramer's rule, we put Eqs. (3.2.1) to (3.2.3) in matrix form.

$$\begin{bmatrix} 3 & -2 & -1 \\ -4 & 7 & -1 \\ 2 & -3 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix} \tag{3.2.6}$$

From this, we obtain

$$v_1 = \frac{\Delta_1}{\Delta}, \qquad v_2 = \frac{\Delta_2}{\Delta}, \qquad v_3 = \frac{\Delta_3}{\Delta}$$

where Δ, Δ_1, Δ_2, and Δ_3 are the determinants to be calculated as follows. As explained in Appendix A, to calculate the determinant of a 3 by 3 matrix, we repeat the first two rows and cross multiply.

$$\Delta = \begin{vmatrix} 3 & -2 & -1 \\ -4 & 7 & -1 \\ 2 & -3 & 1 \end{vmatrix} = \begin{array}{ccc} 3 & -2 & -1 \\ -4 & 7 & -1 \\ 2 & -3 & 1 \\ - & - & - \\ 3 & -2 & -1 \\ - & - & - \\ -4 & 7 & -1 \\ - & - & - \end{array} \begin{array}{c} \\ \\ \\ + \\ + \\ + \end{array}$$

$$= 21 - 12 + 4 + 14 - 9 - 8 = 10$$

Similarly, we obtain

$$\Delta_1 = \begin{array}{ccc} 12 & -2 & -1 \\ 0 & 7 & -1 \\ 0 & -3 & 1 \\ - & - & - \\ 12 & -2 & -1 \\ - & - & - \\ 0 & 7 & -1 \\ - & - & - \end{array} \begin{array}{c} \\ \\ \\ + \\ + \\ + \end{array} = 84 + 0 + 0 - 0 - 36 - 0 = 48$$

$$\Delta_2 = \begin{array}{ccc} 3 & 12 & -1 \\ -4 & 0 & -1 \\ 2 & 0 & 1 \\ - & - & - \\ 3 & 12 & -1 \\ - & - & - \\ -4 & 0 & -1 \\ - & - & - \end{array} \begin{array}{c} \\ \\ \\ + \\ + \\ + \end{array} = 0 + 0 - 24 - 0 - 0 + 48 = 24$$

$$\Delta_3 = \begin{array}{ccc} 3 & -2 & 12 \\ -4 & 7 & 0 \\ 2 & -3 & 0 \\ - & - & - \\ 3 & -2 & 12 \\ - & - & - \\ -4 & 7 & 0 \\ - & - & - \end{array} \begin{array}{c} \\ \\ \\ + \\ + \\ + \end{array} = 0 + 144 + 0 - 168 - 0 - 0 = -24$$

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Thus, we find

$$v_1 = \frac{\Delta_1}{\Delta} = \frac{48}{10} = 4.8 \text{ V}, \qquad v_2 = \frac{\Delta_2}{\Delta} = \frac{24}{10} = 2.4 \text{ V}$$
$$v_3 = \frac{\Delta_3}{\Delta} = \frac{-24}{10} = -2.4 \text{ V}$$

as we obtained with Method 1.

Practice Problem 3.2

Find the voltages at the three nonreference nodes in the circuit of Fig. 3.6.

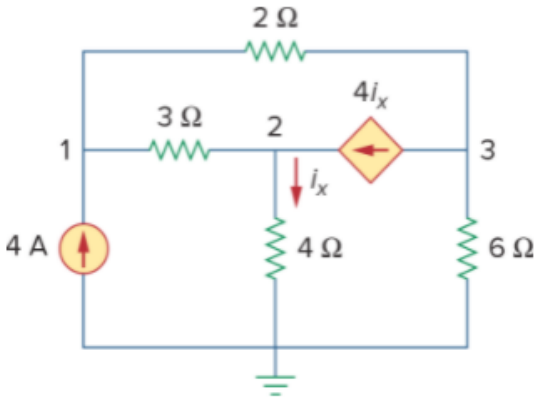
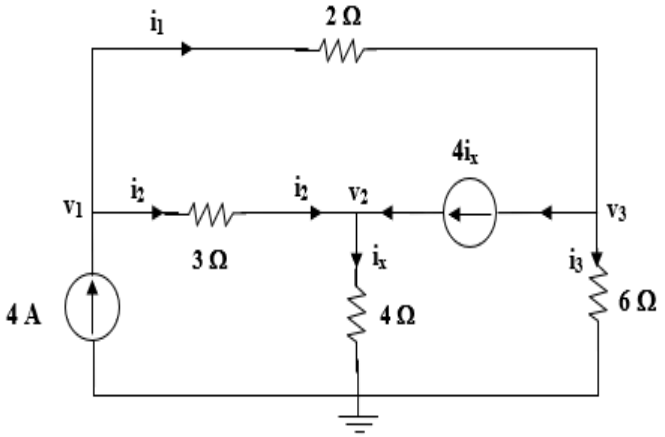


Figure 3.6
For Practice Prob. 3.2.

Answer: $v_1 = 32$ V, $v_2 = -25.6$ V, $v_3 = 62.4$ V.

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At node 1,

$$-4 + i_1 + i_2 = 0 = -4 + \frac{v_1 - v_3}{2} + \frac{v_1 - v_2}{3}$$

or $5v_1 - 2v_2 - 3v_3 = 24$ (1)

At node 2,

$-i_2 + i_x - 4i_x = 0 = -i_2 - 3i_x = 0$ where $i_x = [(v_2 - 0)/4]$ or

$$\frac{v_1 - v_2}{3} + 3\frac{v_2}{4} = 0$$ which leads to $4v_1 + 5v_2 = 0$ (2)

At node 3,

$$-i_1 + i_3 + 4i_x = 0 = \frac{v_3 - v_1}{2} + \frac{v_3 - 0}{6} + 4\frac{v_2}{4}$$

or $-3v_1 + 6v_2 + 4v_3 = 0$ (3)

Solving (1) to (3) gives

$v_1 = \mathbf{32\ V}$, $v_2 = \mathbf{-25.6\ V}$, $v_3 = \mathbf{62.4\ V}$

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Mesh Analysis

Another general procedure for analyzing circuits is to use the **mesh currents** as the circuit variables.

Recall:

- A loop is a closed path with no node passed more than once.
- A mesh is a loop that does not contain any other loop within it.

Mesh analysis uses **KVL** to find unknown currents.

Mesh analysis is also known as *loop analysis* or the *mesh-current method*.

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Mesh Analysis Steps

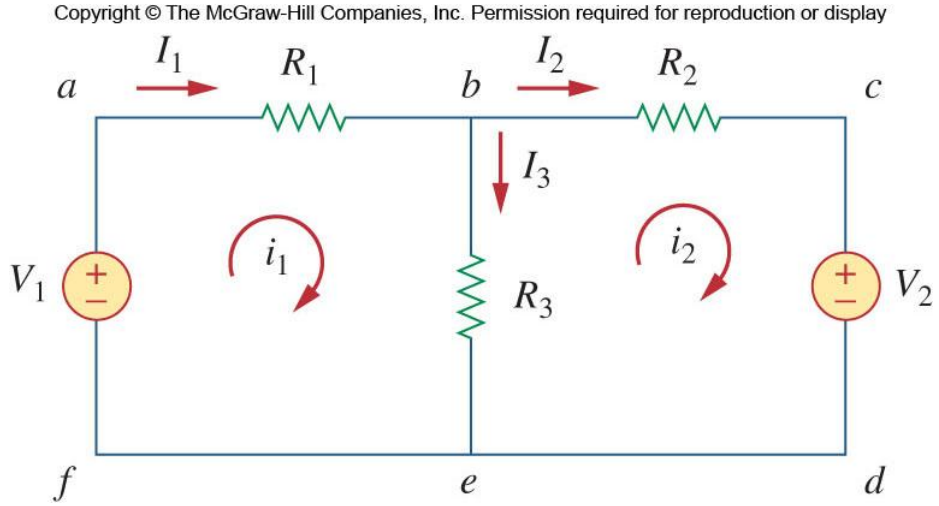
Mesh analysis follows these steps:

1. Assign **mesh currents** $i_1, i_2, \ldots i_n$ to the n meshes.
2. Apply **KVL** to each of the n mesh currents.
3. Solve the resulting ***n* simultaneous** equations to get the mesh currents.

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Mesh Analysis Example

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- The above circuit has **two paths** that are meshes (abefa and bcdeb).
- The outer loop (abcdefa) is a loop, but not a mesh.
- First, mesh currents i_1 and i_2 are assigned to the two meshes.
- Applying **KVL** to the meshes:

$$-V_1 + R_1 i_1 + R_3 (i_1 - i_2) = 0 \quad R_2 i_2 + V_2 + R_3 (i_2 - i_1) = 0$$

$$\Downarrow \qquad\qquad \Downarrow$$

$$(R_1 + R_3) i_1 - R_3 i_2 = V_1 \qquad -R_3 i_1 + (R_2 + R_3) i_2 = -V_2$$

$I_1 = i_1, \quad I_2 = i_2, \quad I_3 = i_1 - i_2$

Example 3.5

For the circuit in Fig. 3.18, find the branch currents I_1, I_2, and I_3 using mesh analysis.

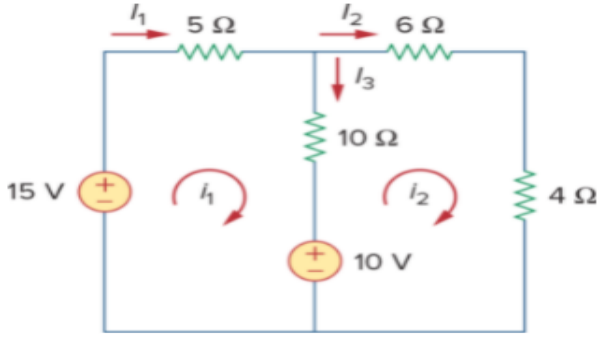


Figure 3.18
For Example 3.5.

Solution:

We first obtain the mesh currents using KVL. For mesh 1,

$$-15 + 5i_1 + 10(i_1 - i_2) + 10 = 0$$

or

$$3i_1 - 2i_2 = 1 \tag{3.5.1}$$

For mesh 2,

$$6i_2 + 4i_2 + 10(i_2 - i_1) - 10 = 0$$

or

$$i_1 = 2i_2 - 1 \tag{3.5.2}$$

■ METHOD 1 Using the substitution method, we substitute Eq. (3.5.2) into Eq. (3.5.1), and write

$$6i_2 - 3 - 2i_2 = 1 \quad \Rightarrow \quad i_2 = 1 \text{ A}$$

From Eq. (3.5.2), $i_1 = 2i_2 - 1 = 2 - 1 = 1$ A. Thus,

$$I_1 = i_1 = 1 \text{ A}, \quad I_2 = i_2 = 1 \text{ A}, \quad I_3 = i_1 - i_2 = 0$$

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■ METHOD 2 To use Cramer's rule, we cast Eqs. (3.5.1) and (3.5.2) in matrix form as

$$\begin{bmatrix} 3 & -2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

We obtain the determinants

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$$\Delta = \begin{vmatrix} 3 & -2 \\ -1 & 2 \end{vmatrix} = 6 - 2 = 4$$

$$\Delta_1 = \begin{vmatrix} 1 & -2 \\ 1 & 2 \end{vmatrix} = 2 + 2 = 4, \quad \Delta_2 = \begin{vmatrix} 3 & 1 \\ -1 & 1 \end{vmatrix} = 3 + 1 = 4$$

Thus,

$$i_1 = \frac{\Delta_1}{\Delta} = 1\ \text{A}, \quad i_2 = \frac{\Delta_2}{\Delta} = 1\ \text{A}$$

as before.

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Example 3.6

Use mesh analysis to find the current I_o in the circuit of Fig. 3.20.

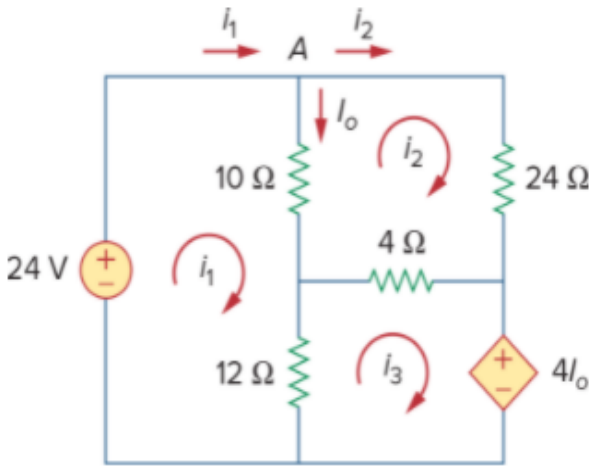


Figure 3.20
For Example 3.6.

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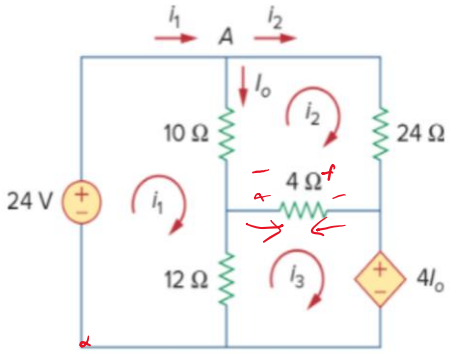
Solution:

We apply KVL to the three meshes in turn. For mesh 1,

$$-24 + 10(i_1 - i_2) + 12(i_1 - i_3) = 0$$

or

$$11i_1 - 5i_2 - 6i_3 = 12 \quad (3.6.1)$$



For mesh 2,

$$24i_2 + 4(i_2 - i_3) + 10(i_2 - i_1) = 0$$

or

$$-5i_1 + 19i_2 - 2i_3 = 0 \quad (3.6.2)$$

For mesh 3,

$$4I_o + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$

But at node A, $I_o = i_1 - i_2$, so that

$$4(i_1 - i_2) + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$

or

$$-i_1 - i_2 + 2i_3 = 0 \quad (3.6.3)$$

In matrix form, Eqs. (3.6.1) to (3.6.3) become

$$\begin{bmatrix} 11 & -5 & -6 \\ -5 & 19 & -2 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix}$$

We obtain the determinants as

$$\Delta = \begin{vmatrix} 11 & -5 & -6 \\ -5 & 19 & -2 \\ -1 & -1 & 2 \end{vmatrix}$$

$$= 418 - 30 - 10 - 114 - 22 - 50 = 192$$

$$\Delta_1 = \begin{vmatrix} 12 & -5 & -6 \\ 0 & 19 & -2 \\ 0 & -1 & 2 \end{vmatrix} = 456 - 24 = 432$$

$$\Delta_2 = \begin{vmatrix} 11 & 12 & -6 \\ -5 & 0 & -2 \\ -1 & 0 & 2 \end{vmatrix} = 24 + 120 = 144$$

$$\Delta_3 = \begin{vmatrix} 11 & -5 & 12 \\ -5 & 19 & 0 \\ -1 & -1 & 0 \end{vmatrix} = 60 + 228 = 288$$

We calculate the mesh currents using Cramer's rule as

$$i_1 = \frac{\Delta_1}{\Delta} = \frac{432}{192} = 2.25 \text{ A}, \quad i_2 = \frac{\Delta_2}{\Delta} = \frac{144}{192} = 0.75 \text{ A},$$

$$i_3 = \frac{\Delta_3}{\Delta} = \frac{288}{192} = 1.5 \text{ A}$$

Thus, $I_o = i_1 - i_2 = 1.5$ A.

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Practice Problem 3.6

Using mesh analysis, find I_o in the circuit of Fig. 3.21.

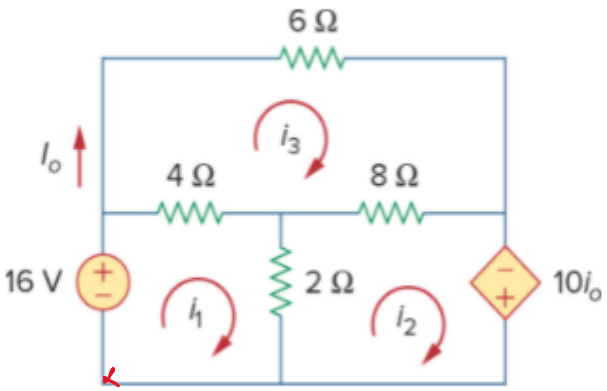


Figure 3.21
For Practice Prob. 3.6.

$I_o = i_3$

Answer: −4 A.

P.P.3.6 For mesh **1**,

$-16 + 4i_1 - 4i_3 + 2i_1 - 2i_2 = 0$

$$-16 + 6i_1 - 2i_2 - 4i_3 = 0 \longrightarrow 3i_1 - i_2 - 2i_3 = 8 \quad (1)$$

For mesh **2**,

$2i_2 - 8i_2$

$$10i_2 - 2i_1 - 8i_3 - 10i_0 = 0 = -i_1 + 5i_2 - 9i_3 \quad (2)$$

But $i_0 = i_3$,

$6i_3 + 8i_3 + 4i_3 - 4i_1 - 8i_2 = 0$

mesh

$$18i_3 - 4i_1 - 8i_2 = 0 \longrightarrow -2i_1 - 4i_2 + 9i_3 = 0 \quad (3)$$

From (1) to (3),

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$$\begin{bmatrix} 3 & -1 & -2 \\ -1 & 5 & -9 \\ -2 & -4 & 9 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 0 \\ 0 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 3 & -1 & -2 \\ -1 & 5 & -9 \\ -2 & -4 & 9 \\ 3 & -1 & -2 \\ -1 & 5 & -9 \end{vmatrix} = 135 - 8 - 18 - 20 - 108 - 9 = -28$$

$$\Delta_1 = \begin{vmatrix} 8 & -1 & -2 \\ 0 & 5 & -9 \\ 0 & -4 & 9 \\ 8 & -1 & -2 \\ 0 & 5 & -9 \end{vmatrix} = 360 - 288 = 72$$

$$\Delta_2 = \begin{vmatrix} 3 & 8 & -2 \\ -1 & 0 & -9 \\ -2 & 0 & 9 \\ 3 & 8 & -2 \\ -1 & 0 & -9 \end{vmatrix} = 144 + 72 = 216$$

$$\Delta_3 = \begin{vmatrix} 3 & -1 & 8 \\ -1 & 5 & 0 \\ -2 & -4 & 0 \\ 3 & -1 & 8 \\ -1 & 5 & 0 \end{vmatrix} = 32 + 80 = 112$$

$$i_1 = \frac{\Delta_1}{\Delta} = \frac{72}{-28} = -2.571, \; i_2 = \frac{\Delta_2}{\Delta} = \frac{216}{-28} = -7.714, \; i_3 = \frac{\Delta_3}{\Delta} = \frac{112}{-28} = -4\text{A}$$

$$I_o = i_3 = \mathbf{-4\ A}$$

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Mesh analysis when…

If the network contains:

- Many series connected elements.
- Voltage sources.
- A circuit with fewer meshes than nodes.

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Nodal analysis if…

If the network contains:

- Many parallel connected elements.
- Current sources.
- Circuits with fewer nodes than meshes.

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